



Homogeneous Propagation Framework: A Comparative Overview

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Abstract

This paper is a clarification of what this framework presently is, and what it is not. The framework recovers a substantial portion of familiar relativistic structure, but it should not be identified either with orthodox special relativity, with orthodox general relativity, or with a finished replacement for them. In the homogeneous regime, contraction, timing structure, and the one-dimensional Lorentz map are recovered in form from a propagation-first construction rather than postulated as primitive spacetime symmetry. In the gravitational regime, weak-field timing, delay, lensing, perihelion precession, and related tests are developed from displacement of the transport-supporting structure rather than from curvature of spacetime itself. The framework is therefore not SR or GR restated in ontology or derivational order, but it still overlaps the tested kinematic and weak-field observational record within its present scope. Those overlaps are real and important, and within those recovered kinematic and weak-field domains the framework is already sufficiently closed to support its present derivations. The empirical divergences from general relativity follow a unifying pattern: the two theories agree wherever both produce finite, bounded results, and they diverge at the loci where GR's solutions produce unbounded quantities — at point sources, at the Schwarzschild radius, at the central singularity. The framework predicts bounded values at each of these loci, and §5.4 consolidates the specific predictions that follow, including a scalar breathing-mode gravitational wave channel derived from the displacement deformation tensor, the replacement of the Schwarzschild horizon with asymptotic time dilation and a finite source-sphere surface, the absence of a photon sphere as a closed-orbit feature, second post-Newtonian divergences from Schwarzschild with calculated magnitudes, and preferred-frame signatures accessible to open-path experiments. The remaining open work enters mainly when the program is pushed

further, particularly toward deeper matter structure, intra-nuclear displacement sourcing from excitation-level theory, and any broader unification with nuclear-scale physics. The focus here is the framework's present status — what has been recovered, what is physically distinct, and where the remaining frontier lies. The central claim is that the framework should presently be read as an alternative derivational and ontological program with significant overlap with SR/GR in its recovered weak-field and kinematic domains, while remaining open in the deeper unificatory sense rather than in those recovered sectors themselves.

Keywords: homogeneous propagation, emergent Lorentz structure, displacement framework, weak-field gravity, ontology, theory status, open dynamics

1. Introduction

A recurring problem for alternative frameworks is that they are often sorted too quickly into the wrong category. If they recover familiar formulas, they are treated as though they must only be a relabeling of established theory. If they depart in ontology, they are often read as claiming more than has actually been shown. This framework has encountered both reactions.

The aim is not to enlarge those claims, but to place them more exactly.

On one side, the kinematic papers recover the standard contraction factor, the timing factor, and the one-dimensional Lorentz transformation in form. The weak-field displacement papers recover the Newtonian limit, the standard weak-field perihelion structure, the logarithmic Shapiro delay, light deflection with the usual coefficient, and a weak-field route toward frame-drag behavior. It is therefore reasonable to ask whether the framework is simply special and general relativity written in different language. That question cannot be answered at the level of category labels alone, because the framework's claim is not merely that it is different from SR/GR, but that it still overlaps the tested kinematic and weak-field observational record while assigning those results a different physical basis.

On the other side, the framework also makes claims that are not standard. It treats the vacuum as a physically structured medium with storage properties, treats Lorentz structure as emergent rather than primitive, and treats gravity as displacement of the transport-supporting structure rather than curvature of spacetime itself. Those are not cosmetic differences. They are differences in what the theory says the world is.

At the same time, the framework is not complete. The displacement-sourcing relation is still

foundational at the present stage: the papers currently treat nuclear-scale excitation dynamics as the source-side bridge rather than deriving that bridge from a finished substrate microphysics. The full microphysical closure is not finished, and the strong-field and fully coupled dynamical sectors remain open.

The clearest reading lies between two familiar simplifications. The framework is not merely SR/GR in different words. But neither is it yet a final replacement theory with every further sector closed and derived.

A more accurate reading is simpler. The framework is best read, at its present stage, as an alternative program with three salient features:

1. broad recovery of standard relativistic structure within the framework's present scope,
2. a distinct underlying physical interpretation of that structure, and
3. explicit remaining gaps that prevent the framework from being treated as complete in a broader unificatory sense.

The aim is to make that status precise. It does not attempt to re-derive every technical result already developed elsewhere. Instead it organizes the existing work around three direct questions: what has already been recovered, in what sense that recovered structure is not just standard relativity again, and what remains genuinely open.

2. The Framework's Present Status in One Sentence

The most concise statement of the framework's present status is the following:

This framework presently recovers large parts of the special-relativistic kinematic structure and the general-relativistic weak-field observational structure, but it does so from a different ontology and derivational order, and it does not yet constitute a final closed theory. Its overlap with the tested kinematic and weak-field record is therefore real without making it identical to SR or GR.

That sentence states the central position.

Three points matter. First, the familiar structures are recovered rather than simply assumed from the start. Second, the physical interpretation assigned to those structures is not the

orthodox one. Third, the framework's remaining openness lies mainly in further sectors, especially those tied to matter structure and intra-nuclear displacement sourcing, rather than in the recovered kinematic and weak-field results themselves.

Those three points organize the discussion that follows.

3. What Has Been Recovered

3.1 Kinematic structure in the homogeneous regime

The homogeneous propagation work begins from a propagation-first condition rather than from an a priori postulate that all inertial frames are fundamentally equivalent. Within that construction, the standard contraction factor and timing factor are recovered from the directional structure of one-way and two-way propagation cost. The one-dimensional Lorentz transformation is then recovered in form from the same propagation geometry. That program should not be collapsed too quickly into a simple "Lorentzian aether" reading, because the framework's treatment of propagation includes directional structure at both transmission and reception rather than only at one end of the exchange; Appendix A explains why that distinction matters for Michelson-Morley-type objections.

This is already a significant result. It means that a structure normally treated as primitive can instead be recovered as the endpoint of a more basic propagation analysis. In the current framework, contraction is not introduced as an independent axiom, nor is time dilation treated as a separate starting law. Both arise from the same interaction-cost structure of motion through the transport-supporting medium.

The framework also extends this recovery program into signal transfer. Doppler structure, aberration, and noncollinear composition are treated as related consequences of the same propagation-based setting rather than as disconnected formula sets imported piecemeal.

3.2 Weak-field gravitational structure

The displacement program then extends the framework beyond the homogeneous regime. Here the working postulate should be read in its narrower source-side form: intra-nuclear action -- the internal dynamics of nuclear-scale excitation structure -- sources displacement of the transport-supporting structure, while the scalar mass-energy assigned to matter is the

downstream bulk measure of that sourcing capacity. Changes in this displacement propagate causally at the medium speed. In the static spherical case, this yields the equilibrium displacement geometry, the cumulative displacement field, and the Newtonian limit.

From that weak-field structure, the framework develops or recovers the following familiar results:

- gravitational clock-rate structure,
- gravitational redshift,
- Shapiro delay,
- Mercury perihelion advance,
- light deflection with the usual weak-field coefficient,
- weak-field observational consistency checks such as lunar laser ranging, and
- a rotating-source extension intended to account for geodetic and frame-drag behavior.

These are not minor overlaps. They cover much of the standard weak-field phenomenology that any serious gravitational framework must meet.

3.3 Why this overlap matters

This overlap should not be minimized. A framework that cannot recover the tested relativistic structures is not competitive. The framework does recover many of them, and that is one reason to take it seriously as a research program.

But overlap in recovered formulas does not by itself settle the question of identity. Two theories may agree over a shared domain while differing in ontology, explanatory order, or extension outside that domain. The next sections therefore ask in what sense the present framework is not simply SR or GR under another name. The structural and ontological arguments are developed in §§4–5; the specific empirical predictions where the framework diverges from GR are consolidated in §5.4.

4. Why This Is Not Simply Special Relativity

4.1 SR takes Lorentz structure as primitive symmetry

In orthodox special relativity, Lorentz invariance is a foundational symmetry statement about spacetime and the laws of physics. The relativity principle and invariant light speed are

primitive ingredients. The Lorentz transformation then expresses the symmetry structure of spacetime itself.

This framework does not start there.

4.2 The program treats Lorentz structure as recovered

In the homogeneous propagation papers, the familiar Lorentz kinematic structure appears as a recovered endpoint of a propagation-first analysis. The framework asks what contraction, timing, and mapping structure follow once propagation through the transport-supporting layer is taken as primary. Lorentz form is then obtained from that more basic construction.

That change in derivational order is significant. In one theory, Lorentz structure is a starting symmetry. In the other, it is an emergent operational structure of systems made from and propagating through a single-speed medium.

4.3 The physical meaning of rods and clocks is different

This difference is not merely mathematical. In the framework's own interpretation, rods and clocks are not primitive ideal objects laid down in an already-given Minkowski spacetime. They are material systems whose internal closure depends on repeated interactions carried by the medium. Contraction and time dilation therefore enter as the self-consistent response of those systems to motion through the transport structure.

Put plainly, the order of explanation is reversed. The framework does not say that matter behaves Lorentz-covariantly because spacetime is Minkowskian in the first instance. It says that matter built from a single-speed propagation medium will realize a Lorentz-type operational structure.

A concrete instance of this distinction appears in the continuous circular-orbit case. The framework's external-field steering integral W_{ext} is the appropriate infinitesimal object for continuous deflection by an external field, and integrating it around a full orbit recovers the standard Thomas precession $2\pi(\gamma - 1)$ at all speeds. The finite Wigner-angle expression for discrete noncollinear boost composition is a different mathematical object, and treating it as if it were the infinitesimal steering integrand produces an orbit sum that misses Thomas precession by a large factor at relativistic speeds — about 3.5% of the correct value at muon $\gamma - 2$ speed. The full derivation of W_{ext} , its external-field-versus-co-moving distinction, and

the explicit step-by-step circular-orbit integration with the muon $g - 2$ numerical comparison are developed in the Wigner rotation paper [3]. The narrow point is that the framework selects a different integrable object than the finite Wigner angle, and that selection is what recovers the correct continuous-orbit limit directly — a concrete example of divergence in mathematical route rather than only in verbal interpretation. The standard Thomas-precession result is itself part of the relativistic baseline; the framework does not contradict it. What the framework does is identify which mathematical object is appropriate to which physical situation, and the difference matters because the external-field case has the deflecting agent operating at the lab clock rate rather than at the moving object's internal clock rate.

4.4 Preferred structure versus undetectable symmetry

The framework also permits a preferred transport-supporting structure at the ontological level, while maintaining that medium-constituted observers recover the familiar relativistic observational structure. This is sharply different from the standard interpretation of SR, where no such preferred underlying structure is admitted.

That does not automatically yield a conflict at the observational level. One can have an observationally emergent Lorentz structure with a deeper preferred substrate if material measuring systems are themselves governed by the same medium. But the explanatory picture is no longer orthodox SR.

4.5 The present conclusion about SR

The point is therefore not that the framework rejects special-relativistic kinematics. Nor is it that it simply restates SR. The more accurate statement is this:

The framework reproduces major special-relativistic kinematic relations while assigning them a different derivational order and a different ontological interpretation.

The framework therefore is not simply SR restated, but a distinct account that overlaps the tested kinematic record from a different physical basis.

5. Why This Is Not Simply General Relativity

5.1 GR identifies gravity with spacetime curvature

In orthodox general relativity, gravity is not a force in the Newtonian sense but the curvature of spacetime itself. The metric is dynamical, the Einstein field equations relate geometry to stress-energy, and free motion follows geodesics of that spacetime geometry.

This framework does not identify gravity that way.

5.2 The program identifies gravity with displacement of transport structure

In the displacement papers, gravity is treated as displacement of the transport-supporting spatial structure. The framework's older shorthand was not 'matter tells spacetime how to curve,' but rather 'mass tells space how to displace, and displaced space tells mass how to move.' The sharper source-side version is narrower: nuclear-scale excitation dynamics tell transport-supporting space how to displace, and displaced transport tells matter how to move. That is a different ontology, even where the resulting weak-field coordinate-level structure overlaps standard GR.

The framework does develop an effective coordinate geometry. But this effective geometry is downstream of the displacement picture; it is not the primitive ontological object. On the present reading, geometry here is descriptive of the displaced state, not the fundamental thing being displaced. The time and radial components are modified by the displacement field, while the angular sector remains Euclidean in the present weak-field construction. That is not merely a restatement of GR's geometrical foundation.

5.3 Agreement in the weak field does not imply identity

It should also be stated carefully that weak-field agreement with standard tests does not make the theory identical to GR. The same observational structure may arise from more than one underlying picture over a restricted domain. The framework's Mercury, Shapiro, and light-deflection results are important consistency checks, but they do not erase the distinction between curvature ontology and displacement ontology.

5.4 Where the predictions actually diverge

Ontological distinctness without empirical distinctness is a weak position. The structural and explanatory differences developed in §§5.1–5.3 would carry less force if the framework's predictions tracked GR's predictions identically at every accessible precision level. They do not. The framework's internal logic produces specific divergences from general relativity, and those divergences follow a unifying pattern that is worth stating before the specific items.

The unifying pattern. The framework and general relativity agree wherever both theories produce finite, bounded results. They diverge at the loci where GR's solutions produce unbounded quantities. At a point source, GR's force diverges; the framework treats every source as an extended displacement region with force going to zero at the center. At the Schwarzschild radius, GR's coordinate quantities diverge and an event horizon appears; the framework produces asymptotic time dilation and a finite source-sphere surface, with no horizon. At the central singularity, GR's curvature diverges; the framework has no singularity to diverge at. The empirical divergences from GR are therefore not a scattered list of corrections in different regimes. They are a single structural fact viewed at different scales: GR's solutions produce infinities at certain loci, and the framework's solutions do not.

Two tables make this concrete. The first compares GR's point-mass idealization with the framework's reading of an ordinary extended object whose interior consists of summed displacements from its constituent matter — a planet, a star, a nucleus treated as composite. The second compares GR's point-mass idealization with the framework's reading of what GR would call a black hole — a solid displacement region treated as an irreducible source rather than as a collection of summed displacements.

Table 1: GR point mass versus framework summed-displacement object (ordinary matter — Sun, planet, ordinary nucleus)

Quantity	GR — Point mass ($r \rightarrow 0$)	Framework — summed- displacement interior
Force $ \nabla \Sigma $	$\rightarrow \infty$	$\rightarrow 0$ (linear toward center)
Cumulative displacement Σ	$\rightarrow \infty$	$3GM/2c^2R \approx 3.2 \times 10^{-6}$ (Sun)
Time dilation	$\rightarrow \infty$	$\approx 1 - 3.2 \times 10^{-6}$ (Sun)

Quantity	GR — Point mass ($r \rightarrow 0$)	Framework — summed- displacement interior
Proper velocity stretch	$\rightarrow \infty$	parts per million above c
Light-crossing transit time	$\rightarrow 0$	slightly stretched
Escape velocity V_{esc}	$\rightarrow \infty$	$\ll c$
Transport space	normal	normal, slightly dilated

Table 2: GR point mass versus framework solid-displacement object (irreducible source — black-hole region, fundamental nuclear displacement)

Quantity	GR — Point mass ($r \rightarrow 0$)	Framework — solid-displacement interior
Force $ \nabla \Sigma $	$\rightarrow \infty$	peaks at R_s , $\rightarrow 0$ at center
Cumulative displacement Σ	$\rightarrow \infty$	bounded $[1/2, 3/4]$
Time dilation	$\rightarrow \infty$	$[2/3, 4/7]$ — finite
Proper velocity stretch	$\rightarrow \infty$	$[1.5c, 1.75c]$ proper
Light-crossing transit time	$\rightarrow 0$	scales linearly with R_s
Escape velocity V_{esc}	$\rightarrow \infty$	$[c, \sqrt{3/2} c]$
Transport space	normal	not well defined inside

The two tables share a structural observation: every quantity GR sends to infinity, the framework returns as a bounded value. The two differ in what they say about the interior, because they describe physically different objects. A summed-displacement interior is ordinary matter — its transport structure is well-defined, light propagates normally, and the surface densities and central displacements are extremely small in the relevant regimes. A solid-displacement interior is a single irreducible source — its bounded values are computed from

the framework's geometric construction directly, and the transport structure is not well-defined within the source-sphere because the source-sphere is not made of summed transport elements.

With that unifying picture in place, the specific predictions follow. They are listed below in two groups by accessibility.

Predictions accessible in current or near-term data.

Breathing-mode scalar radiation in gravitational waves. The framework's gravitational waves are longitudinal compression waves in the transport structure rather than transverse-traceless tensor modes. For eccentric binaries this produces an additional scalar radiation channel — a breathing mode — that is absent in general relativity. The eccentricity enhancement function $f(e)$ differs from the GR expression by an amount derived from the displacement deformation tensor, with additional power at the radial oscillation frequency. Developed in §§9.5–9.6 of the weak-field observational tests paper.

Open-path clock-transport asymmetry. Closed-path clock-transport experiments such as Hafele-Keating do not distinguish the framework from GR because the preferred-frame cross-term cancels exactly when the clock returns to its starting point. An open-path clock-transport experiment — one in which the clock does not return — would retain a cross-term between the preferred-frame velocity (the solar-system CMB-dipole velocity of approximately 370 km/s) and the local transport velocity. The required precision is extreme, but the prediction is structurally definite. Developed in §3.5 of the weak-field observational tests paper.

CMB-aligned well-depth asymmetry in orbital eccentricity. Because the solar system moves through the transport structure at approximately 370 km/s, the retarded displacement field of the Sun is not perfectly symmetric. An orbiting body sees a slightly shallower effective well on the leading edge of its orbit and a slightly deeper well on the trailing edge. This produces a CMB-aligned modulation of orbital eccentricity, distinct from the solar tidal modulation aligned with Earth's perihelion direction. The two directions differ by approximately 67° in ecliptic longitude and are in principle resolvable in the lunar laser ranging dataset after subtraction of the solar tidal contribution. Developed in §8.4 of the weak-field observational tests paper.

Predictions below current thresholds but structurally definite.

2PN divergence from Schwarzschild. The effective coordinate geometry of the displacement framework matches Schwarzschild at first post-Newtonian order. At second order the two

diverge. The framework's exact metric components — $f = 1/(1 + \Sigma)^2$ from the clock-rate relation, $g = D^2$ from the radial deformation, and $h = 1$ from coordinate anchoring of $F \cdot H$ — produce specific corrections at σ^2 order. For Mercury, the 2PN correction is approximately 0.002 milliarcsec per century. For the Hulse-Taylor binary it reaches roughly 0.03 arcsec per year. For light propagation, a $\frac{7}{8}\sigma^2$ correction modifies the radial light speed at second order, with consequences for high-precision Shapiro delay and bending-angle measurements. None of these is currently testable, but all follow from the framework's internal logic without adjustable parameters. Developed in Appendix C of the displacement dynamics paper.

No Schwarzschild horizon. The framework's strong-field structure replaces the Schwarzschild event horizon with asymptotic time dilation and a finite source-sphere surface, as shown in Table 2 above. The radial deformation D remains exactly linear in σ , and the displacement equilibrium develops a finite source-sphere rather than a coordinate singularity. This is a strong-field prediction; its observational consequences for compact-object phenomenology — accretion signatures, merger waveform tails, the apparent shadow of compact objects — remain open. Developed in Appendix B of the displacement dynamics paper.

No photon sphere as a closed-orbit feature. In GR the Schwarzschild geometry admits unstable circular photon orbits at $r = 3GM/c^2 = 1.5 r_s$, forming the photon sphere. The framework's transport structure does not support closed photon orbits at any radius: light propagates along straight-line geodesics through the displaced transport, and the geometry produces bending without admitting orbital closure. This is structurally distinct from the no-horizon prediction — the photon sphere sits well outside any horizon and is a feature of GR's geometric closure, not its singular structure — and represents a separate observational signature for high-resolution imaging of compact objects.

Modified mass composition and binding-energy structure. Mass composition in the framework follows from displacement support rather than from energy bookkeeping alone. In the weak-overlap regime the composition law is additive in the displacement parameter d^2 , giving the standard result; in regimes where bound systems share displacement support, the binding energy is interpreted as shared support rather than as subtracted rest energy. This affects the framework's predictions for black hole mergers, where the merged source's displacement support is set by volume addition rather than by energy summation. Developed in Appendix D of the displacement dynamics paper.

The point of cataloguing these together is to forestall the conclusion that the framework is empirically indistinguishable from GR. It is not. Within the present scope it agrees with the

tested record, but it makes specific further predictions, some accessible in near-term data and others accessible in principle if not in current practice. The unifying observation is that GR's predictions and the framework's predictions are different precisely where GR's solutions produce infinities — and not by accident, but because the framework's construction does not allow such infinities to arise in the first place.

5.5 The dynamical structure is organized differently

A further difference concerns how the gravitational sector is closed. General relativity closes its gravitational sector through a single tensorial field equation whose strong-field solutions admit singularities and event horizons. The framework closes its gravitational sector through a two-layer coupled structure: the retarded displacement map for the cumulative displacement field, the effective coordinate-level geometry that follows from the displacement, and the orbit and propagation equations on that geometry. Weak-field tests, lensing, perihelion advance, Shapiro delay, frame drag, and gravitational radiation are developed within this structure without additional postulates beyond the displacement-sourcing relation.

The comparison of formal completeness should therefore not be stated as "framework less complete than GR." The framework's gravitational sector is closed in its own terms over the same observational regimes GR is closed, and it remains finite in regimes where GR's solutions produce singular quantities. The structural difference is in the choice of formal vehicle. GR collapses the geometry and its dynamics into one equation, which produces tensorial elegance but also produces the singularities that come with that closure. The framework separates the displacement field and the effective geometry into coupled layers, which produces no such elegance but also produces no such singularities. Neither closure is incomplete; they are differently organized.

What does remain genuinely open in the framework is the deeper microphysical account of why intra-nuclear action has the sourcing role assigned to it. That open question lies outside the gravitational sector proper. General relativity does not answer it either; GR takes the stress-energy tensor as input and does not derive its source. The framework's source-side problem is therefore a different problem than the one GR has solved, and the framework's lack of a finished answer to it does not constitute incompleteness *in the gravitational sector* in the sense in which GR is complete in that sector.

5.6 The present conclusion about GR

The point can therefore be stated as follows:

The framework reproduces substantial weak-field gravitational structure associated with GR, but it does not identify gravity with curved spacetime in the orthodox sense, it organizes its dynamical structure into two coupled layers rather than into a single tensorial field equation, and it produces bounded values at the loci where GR's solutions produce infinities. It is therefore not a less complete version of GR but a differently organized account of the same observational regimes, with specific further predictions where GR's solutions become singular.

The framework therefore is not simply GR restated, but a distinct account that overlaps the tested weak-field observational record from a different physical basis.

6. The Importance of Distinguishing Recovery from Reduction

A recurring ambiguity in theory comparison concerns the difference between **recovering** a structure and **reducing** one theory to another.

To recover a structure is to derive a formula, relation, or observable pattern that matches a known theory in a certain regime. To reduce one theory to another would be to show that the first is simply the second, perhaps in altered notation, with no additional ontological or explanatory content.

The framework currently supports the first claim in many places and does not support the second.

It recovers Lorentz-form kinematics. It recovers important weak-field gravitational observables. But the framework does not reduce itself to standard SR/GR because:

1. it starts from different postulates and a different explanatory hierarchy,
2. it gives a different physical interpretation of rods, clocks, and gravitational structure,
3. it allows a substrate / transport ontology not admitted by orthodox relativity, and
4. it remains incomplete in places where SR/GR are already formally closed.

This distinction matters both rhetorically and technically. Overstating the case as 'this is just relativity again' erases the framework's actual content. Overstating it as 'this replaces relativity in finished form' erases its actual incompleteness. The correct middle statement is stronger than either caricature.

7. What Remains Open

The framework is not unfinished in any trivial sense. Naming the remaining open work should not obscure the fact that the recovered kinematic and weak-field sectors already overlap the tested record within the framework's present scope. The open items are not merely cosmetic or pedagogical. They concern foundational closure.

7.1 The displacement sourcing relation remains foundational

At present, the framework should not be summarized merely as "mass-energy sources displacement." That phrase is useful as a bulk shorthand, but it makes mass sound like an unexplained primitive. The more precise source-side statement is that intra-nuclear action -- the internal dynamics of nuclear-scale excitation structure -- sources displacement of the transport-supporting structure. The scalar mass-energy assigned to matter is then the downstream bulk bookkeeping of that sourcing capacity.

This sourcing relation remains foundational in the sense that it has not yet been reduced to a complete excitation-level or nuclear-scale account internal to the broader program. Within the recovered kinematic and weak-field gravitational domain, the framework is already operationally closed enough to produce the main derived results. The open question is not whether that sector functions, but whether intra-nuclear displacement sourcing can be derived from a deeper matter theory rather than retained as a primitive bridge.

This is the most important incompleteness to state plainly.

7.2 The full microphysical closure is not complete

The broader program aims to ground matter, propagation, and gravity in a common medium picture. But the derivation from substrate to stable excitation to intra-nuclear action to bulk mass-loading to displacement is not yet fully closed. The excitation-layer work points in that

direction, but the chain is not complete enough to claim that the displacement source sector has been wholly derived from deeper first principles internal to the framework.

That distinction matters. This paper should not imply that the recovered SR/GR-level structure is somehow still waiting to be closed on its own terms. Rather, the open work begins when the framework attempts to unify that recovered structure with a deeper account of matter, excitation, intra-nuclear action, and nuclear-scale behavior.

7.3 Strong-field and fully coupled regimes remain frontier problems

Most of the present success lies in homogeneous and weak-field sectors. Strong-field behavior, genuinely nonlinear multi-body coupling, compact-object dynamics, and full self-consistent source-field backreaction remain frontier tasks rather than settled accomplishments.

7.4 Rapid-rotation regime and closed field-law remain frontier

The rotating-source vector sector is developed for the weak-field slow-rotation regime. The vector response has a derivation from the spin-odd azimuthal motion of displacement elements, the slow-rotation effective metric is constructed in Appendix E of the displacement dynamics paper, and the resulting frame-drag estimate lies near the observed Gravity Probe B signal; the geodetic branch follows directly from the effective coordinate geometry and is treated as mature. What remains frontier work is the rapid-rotation regime, where the slow-rotation expansion no longer suffices, and the formal closure of the field law that would determine the vector response from a deeper action principle.

7.5 Why naming the open problems helps rather than hurts

A theory gains credibility by marking its unresolved sectors accurately. Here, naming the open problems does not weaken the recovered results. It sharpens them. It makes clear which claims are already supported, which remain programmatic, and where the next technical work belongs.

8. What the Framework Should Claim at the Present Stage

The framework should presently make six claims, and no stronger claim than these is needed.

8.1 First claim: recovered kinematic structure

It can claim that the homogeneous propagation program recovers the familiar contraction, timing, and Lorentz-form kinematic relations from a propagation-first starting point.

8.2 Second claim: recovered weak-field gravitational structure

It can claim that the displacement program recovers the standard weak-field observational structure across several major tests, including timing, delay, orbital precession, and lensing. That overlap should be stated as overlap with the tested record, not as identity of ontology or causal assignment with GR.

8.3 Third claim: distinct ontology

It can claim that these results arise within a framework whose ontology differs from orthodox relativity: a structured vacuum / transport medium, emergent rather than primitive Lorentz structure, and gravity as displacement rather than as fundamental spacetime curvature. The point is not merely that the framework is not SR/GR, but that this ontological difference still leaves real overlap with the observed kinematic and weak-field benchmarks in the domains already treated.

8.4 Fourth claim: specific predictions that diverge from GR

It can claim that beyond the recovered weak-field overlap, the framework makes specific empirical predictions that diverge from general relativity, following a unifying pattern: the two theories agree where both produce finite, bounded results, and they diverge at the loci where GR's solutions produce unbounded quantities. The specific predictions include the breathing-mode scalar radiation channel in eccentric binary inspirals, the replacement of the Schwarzschild horizon by asymptotic time dilation and a finite source-sphere, the absence of a photon sphere as a closed-orbit feature, second post-Newtonian divergences from

Schwarzschild with calculated magnitudes, preferred-frame signatures retained by open-path or directional experiments, and the CMB-aligned modulation of orbital eccentricity. These follow from the framework's internal construction without adjustable parameters and are catalogued in §5.4 with pointers into the technical papers.

8.5 Fifth claim: incompleteness is acknowledged

It can claim that the framework is not yet a closed final theory in the broader unificatory sense. The recovered SR/GR-level kinematic and weak-field sectors are already sufficiently closed to support the present derivations. What remains postulated, provisional, or under development are the deeper intra-nuclear sourcing and excitation-level sectors needed if the framework is to absorb nuclear theory and related structure into the same program.

8.6 Sixth claim: the value of the program lies in explanatory reorganization

It can claim that the framework's value is not only in reproducing familiar formulas, but in reorganizing them into a different explanatory hierarchy that may connect kinematics, matter structure, and gravity more tightly than the standard compartmentalized picture.

These claims are already strong and do not require rhetorical inflation.

9. Misreadings to Avoid

A paper of this kind should also pre-empt several foreseeable misreadings.

9.1 Not 'SR/GR are wrong, therefore this wins'

That is not the present paper's claim. The framework is strongest where it reproduces known successful structure, not where it merely negates it.

9.2 Not 'this is only semantics'

The framework does not differ from SR/GR only in terminology. It changes the explanatory direction and physical ontology in substantive ways.

9.3 Not 'recovered weak-field overlap means full equivalence'

Agreement in the weak-field regime does not establish identity with GR, particularly when the ontology and unfinished sectors differ.

9.4 Not 'open problems mean the recovered results do not count'

Incomplete closure does not nullify already-derived overlap. It only limits how far the present claims may be extended.

9.5 Not 'just another singularity-resolution program'

The framework is not alone in being dissatisfied with general relativity's singularities. Edge and boundary theories — including the holographic-principle lineage from 't Hooft and Susskind, AdS/CFT, and more recent work on edge modes and asymptotic boundaries in gravitational physics — also seek to resolve GR's singular structure. A reader familiar with that landscape will reasonably ask whether the framework is another entry in the same class. It is not, and the distinction is worth stating.

Edge and boundary approaches typically resolve singularities by relocating the relevant degrees of freedom to a codimension-one boundary. The mathematical singularity remains, but it is reinterpreted as the limit of a different structure — information is held on a holographic boundary, or singular interior degrees of freedom are absorbed into edge modes. The framework's resolution is structurally different: there is no point-mass idealization to begin with. Every source is an extended displacement region whose interior is finite and well-defined, and the force gradient goes to zero at the center rather than diverging there. The resolution is constitutive, not interpretive.

Feature	Edge / boundary singularity-resolution	The present framework
Singularities	Resolved via edge modes or holographic bounds; the mathematical singularity is	Resolved by replacing point-mass idealization with extended displacement; force $\rightarrow 0$ at the center

Feature	Edge / boundary singularity-resolution	The present framework
	reinterpreted	constitutively
The "edge"	A mathematical boundary where information is held	A physical transition in the displacement of the transport medium
Galactic scale	Boundary conditions typically ignore the medium	Outer edge of the displacement profile recovers flat rotation curves without invoking dark matter
Force	Defined by curvature near the boundary	Defined by the gradient $ \nabla \Sigma $ of cumulative displacement

The galactic-scale entry deserves its own remark. Flat rotation curves are recovered from the framework's occupancy and displacement structure without invoking dark matter. The recovery uses a constrained radial occupancy profile that is effectively a single declining gradient control: central concentration, broad disk, outer transition, and faint tail are not independent fit parameters but ordered, monotonic samples of one underlying gradient function. The velocity response of that profile produces the observed long-flat shape and returns surface densities in solar masses per square parsec that are within the physically reasonable range for the relevant galaxy types. An interactive simulator implements the full chain from photometric profile through displacement field to velocity curve. The systematic multi-galaxy validation, including comparison against the standard dark-matter-halo fits used in the SPARC literature, is in progress; the validation pipeline must work against radial photometry rather than against Sérsic-fit intermediate brightness columns, since the rim-transition features the model treats as physically important are smoothed away by Sérsic preprocessing. A companion paper developing the methodology and the multi-galaxy fits is in preparation.

The point is therefore not that the framework competes with edge or boundary programs on those programs' own terms. The point is that the framework dissolves the problem at a different level. Edge programs ask how to interpret an idealization that produces unbounded quantities. The framework does not adopt the idealization in the first place, and therefore does not need to interpret what to do about the infinities it would otherwise produce.

10. A Useful Comparison Table

For clarity, the framework may be summarized in the following comparative form.

Topic	Orthodox SR/GR reading	Framework reading
Light speed / propagation	Primitive relativistic postulate or spacetime property	Consequence of the medium's propagation-supporting structure
Lorentz structure	Fundamental spacetime symmetry	Emergent operational structure recovered from propagation
Rods and clocks	Primitive measuring systems in relativistic spacetime	Material closures sustained by the same medium structure
Gravity	Curvature of spacetime	Displacement of the transport- supporting structure
Weak-field observables	Consequences of metric curvature	Consequences of cumulative displacement and effective geometry
Behavior at singular loci	Produces infinities at point sources, the Schwarzschild radius, and the central singularity	Produces bounded values at every locus where GR diverges
Gravitational wave structure	Transverse-traceless tensor modes only	Longitudinal compression waves with a scalar breathing mode for eccentric sources
Strong-field interior structure	Schwarzschild event horizon and singularity	Asymptotic time dilation with a finite source-sphere surface
Photon sphere	Closed photon orbits at $1.5 r_s$	No closed photon orbits at any radius
2PN comparison	Schwarzschild result	Specific divergence (e.g. 0.03 arcsec/yr for Hulse-Taylor)

Topic	Orthodox SR/GR reading	Framework reading
Ontology	No preferred medium required	Preferred transport-supporting structure permitted
Preferred-frame signatures	None at any order	Cancel in closed-path experiments; present in open-path or directional experiments
Formal closure	Closed via single tensorial field equation; singularities accepted as a consequence	Closed via two coupled layers (displacement field plus effective geometry); finite throughout

The table is not meant as a proof. It is a compact statement of the present difference in explanatory order and status.

11. Why This Clarification Matters for the Paper Series

The paper series benefits from this clarification in at least three ways.

First, it tells readers what sort of work they are reading. They are not being asked to choose immediately between total equivalence and total replacement. They are being asked to evaluate a structured alternative program that overlaps strongly with standard relativity where it should.

Second, it protects the stronger technical papers from being judged by a distorted standard. A paper that recovers kinematic form from a new starting point should not be criticized for not also being a full gravitational microphysical theory. A weak-field gravity paper should not be criticized for not yet closing the excitation ontology. Distinct tasks belong to distinct papers.

Third, it gives the program a cleaner research trajectory. The next work is not 'prove everything all at once.' It is to close the specific remaining gap: derive intra-nuclear displacement sourcing more deeply. The rotating-source vector sector still has a rapid-rotation regime and a formally closed field law to develop, but its weak-field slow-rotation behavior is in hand. The genuinely open path is the source-side microphysics.

12. Conclusion

The homogeneous propagation framework should presently be understood neither as a mere relabeling of special and general relativity nor as a finished replacement for them. It has already done more than propose a speculative vocabulary. It has recovered significant kinematic and weak-field gravitational structure from a different starting point. But it has also not yet earned the right to be described as fully closed.

That intermediate status is not something to conceal. It is the correct description of the framework's present maturity, and it is a stronger position than either of the two easy misreadings. The framework is not empty because it overlaps relativity. It is not complete because it has strong results. Both statements can be true at once.

The strongest present statement is therefore also the most disciplined one: the framework recovers substantial relativistic structure in form and observation, interprets that structure through a distinct medium-and-displacement ontology, and remains unfinished at the deeper dynamical and substrate-closure level. Read that way, it is neither 'just SR/GR again' nor 'already complete beyond SR/GR,' but an alternative program whose overlap, differences, and open problems can all be stated plainly.

Acknowledgment of Status

The purpose here is clarification of the framework's present state. It does not supersede the technical derivations in the companion papers, but states how they should presently be read together.

Appendix A. Michelson-Morley, Transmission Aberration, and Why the Framework Is Not Simple Lorentzian Aether

A likely objection to any framework with a preferred underlying transport structure is that it sounds like a return to Lorentzian aether. If that were all the present framework amounted to, then the obvious historical objection would follow immediately: simple emission through a preferred medium should generically expose absolute motion through Michelson-Morley-type

optical tests. That objection matters and should be addressed directly.

The distinction advanced here is that the present framework does not treat optical propagation as a one-sided effect in which only the receiving frame contributes the relevant aberrational response. Instead, the broader propagation program treats directional effects as entering through both transmission and reception. In the present language, the emitted propagation direction is not independent of the state of the transmitting system, and the received direction is not independent of the state of the receiving system. That two-sided structure is what separates the framework from a naive Lorentzian-aether reading.

The Michelson-Morley issue can be stated at a high level as follows. In a simple preferred-medium picture, one expects an arm aligned with the absolute-motion direction to incur a different optical cost from an arm transverse to it, producing a second-order fringe shift under rotation. Standard Lorentzian rescue arguments then rely on compensating contraction and timing effects to suppress the observable difference. The present framework proposes a different route. Because propagation direction is treated as conditioned by the transmitting structure as well as by the receiving structure, the optical path comparison is not the same as in a one-sided ballistic or one-sided medium model. The relevant directional normalization already enters at launch, not only at detection.

This matters even in the near-frame or small-separation case. If absolute motion is present, then a purely reception-only aberration picture still leaves an asymmetry in how the launched beam samples the longitudinal and transverse arms. By contrast, if transmission already carries the corresponding directional adjustment, then the outbound and return geometry are altered before the final reception event is considered. In that case the Michelson-Morley null result is no longer being explained by the same mechanism usually associated with simple Lorentzian aether.

The full derivation is developed in Appendix C of the light propagation paper [1]. There, the moving interferometer is constructed from three ingredients that all follow from the propagation-first analysis: longitudinal arm contraction by $\lambda(\beta)$, signed planar aberration applied at each launch event (from Appendix B of the same paper), and leg lengths fixed by the solved native propagation delay rather than imposed from the contracted arm offsets. Under those three ingredients, the two-way coordinate transit times along the vertical and horizontal arms are equal, and the local two-way clock times are equal as a direct consequence. The null result is therefore produced by the framework's own construction without being declared as an axiom. The interferometer cannot use its own internal two-way

optical structure to reveal its uniform motion through the medium because rods, clocks, and signal paths are all built from the same single-speed medium and co-adjust together.

The role of the present appendix is therefore not to redo that derivation but to mark the exact conceptual difference that §3.1 depends on. The present framework is not merely saying that matter contracts in a preferred frame and therefore hides the aether. It is saying that the propagation map itself is different because both emission and reception are medium-conditioned processes. That is a materially different claim, and the math behind it is the appendix in [1].

Accordingly, the correct comparison is not between the present framework and a textbook caricature of aether theory, but between two different propagation models: one in which the optical effect is effectively one-sided, and one in which directional structure is built into both ends of the exchange. If the latter is the correct description, then Michelson-Morley no longer functions as a simple argument that any preferred underlying structure must reduce to the historically rejected form.

For present purposes, that point is sufficient. The role of the appendix is to keep the reader from collapsing the framework into the phrase "Lorentzian aether" without noticing that the optical mechanism being proposed is different at the level of propagation itself.

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